

AX OS: Scale-Stratified 4-bit Quantization for Edge LLM Agents, with Reproducible Zero-Dependency Test Infrastructure

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Abstract

We study scale-stratified 4-bit quantization for edge LLM deployment using AX OS, a TypeScript multi-agent library with reproducible test infrastructure. **Primary finding:** standard WikiText-2 evaluation across three Qwen2.5 sizes (1.5B, 3B, 7B) and Mistral-7B shows that within the Qwen family, uniform q4 (group size 64) perplexity degradation is nearly flat at +15.0%, +11.7%, +13.5%—scale alone does not drive quantization robustness. Cross-architecture variation dominates: Mistral-7B degrades only +3.9% versus Qwen2.5-7B’s +13.5%–+16.3% at the same $3.5\times$ compression ratio, a $3.5\times$ – $4.2\times$ gap that motivates architecture-stratified rather than scale-only quantization decisions. Mixed-precision alternatives (3-bit low layers) collapse in quality at 1.5B scale, reinforcing the uniform scheme choice. To ensure these results are exactly reproducible on any target machine, we contribute two pieces of deployment infrastructure: **(1) a zero-dependency `_harness.mjs` shim** backed entirely by Node.js built-ins (`node:test`, `node:assert/strict`), achieving 27/27 test pass without `npm install`; and **(2) inline oracle switching** via a single `BRAIN_REAL=1` environment variable that activates a live Python subprocess oracle without a compiled build artifact. The compressed Mistral-7B artifact is registered in the `AEQIntegration` model registry and validated by the installation-free harness, making it directly loadable as a drop-in agent model.

1 Introduction

Deploying large language model (LLM) agents on edge hardware requires first answering a model-selection question: which model, at which precision, retains acceptable quality under memory constraints? Answering this question reproducibly requires test infrastructure that survives clean-environment deployments and oracle integration that couples the agent to external simulators without heavyweight middleware. Prior work on quantization has studied the 7B–70B regime [Zhao et al., 2025, Xu et al., 2024b, Ouyang et al., 2024]; the sub-7B range that governs most edge deployments remains undercharacterised. Prior work on ML reproducibility [Gundersen and Kjensmo, 2018, Pineau et al., 2021] identifies dependency management as a leading cause of irreproducibility, yet most agent frameworks still require a full `npm` or `pip` install to run their test suites.

The dominant post-training quantization (PTQ) methods—GPTQ [Frantar et al., 2023], AWQ [Lin et al., 2023], and SqueezeLLM [Kim et al., 2024]—demonstrate that 4-bit weight quantization can be applied with minimal quality loss on billion-parameter models. Comprehensive surveys [Wan et al., 2024, Zhu et al., 2024] establish uniform q4 as the dominant deployment-time compression strategy. Existing scaling analyses [Xu et al., 2024b, Ouyang et al., 2024] cover the 7B–70B range; we extend the measurement below 7B to the sizes that govern most edge deployments.

Once compressed, agents must be validated on the same hardware they target. Production local-inference studies [Rajesh et al., 2025] show that frame-

work choice among llama.cpp, MLX, MLC-LLM, Olama, and PyTorch MPS creates measurable differences on Apple Silicon. Yet validating behavior across these platforms requires a test harness that runs without a full dependency install—a gap AX OS addresses with a Node.js built-in shim—and an oracle coupling mechanism that does not depend on a compiled build artifact.

This paper makes one empirical contribution supported by two infrastructure contributions:

1. **Scale-stratified 4-bit quantization measurements (primary).** WikiText-2 standard evaluation across three Qwen2.5 sizes and Mistral-7B shows that within the Qwen family, uniform q4 Δ PPL is nearly flat (+15.0% at 1.5B, +11.7% at 3B, +13.5% at 7B—uniform local-pipeline), while Mistral-7B degrades only +3.9%—suggesting that architecture is a stronger predictor than scale for quantization robustness at constant 3.5 $\times$ compression, within this two-family comparison.
2. **Zero-dependency node:test harness (enabling).** A 40-line shim backed by Node.js built-ins achieves 27/27 test pass on any clean Node.js ≥ 18 installation without `npm install`, including the tests that validate the compressed-model registry.

Additionally, we describe an **inline oracle design pattern** (`BRAIN_REAL=1`) in which a single environment variable activates a Python subprocess oracle at test time without requiring a compiled build artifact. This pattern is structurally verified in the AX OS codebase but not executed against the live oracle in this work, and is offered as an engineering reference rather than an empirical finding.

2 Related Work

2.1 Post-Training Quantization for LLMs

Post-training quantization (PTQ) has become the dominant compression strategy for LLMs because it

avoids retraining costs while yielding practical inference speedups [Wan et al., 2024]. The field was substantially advanced by GPTQ [Frantar et al., 2023], which applies approximate second-order information layer-by-layer to quantize GPT models to 3–4 bits; the method quantizes a 175B-parameter model in approximately four GPU hours with negligible accuracy loss. AWQ [Lin et al., 2023] observes that only 1% of weights are salient (as measured by activation magnitudes) and protects those weights with higher precision, achieving over 3 $\times$ end-to-end speedup on edge GPUs while winning the MLSys 2024 Best Paper Award. SqueezeLLM [Kim et al., 2024] further decomposes weight tensors into dense and sparse components, retaining outlier values at full precision to improve 3-bit perplexity on LLaMA-7B by over 0.3 points on C4.

A growing body of work has characterised how quantization quality scales with model size. Zhao et al. [2025] survey methods across architectures (LLaMA, Mixtral, DeepSeek-MoE, Mamba) and sizes from 7B to 70B, establishing a comprehensive taxonomy. Xu et al. [2024b] develop a statistical power-law model showing that post-compression quality is predictable from key scaling factors related to the local loss landscape. Ouyang et al. [2024] examine over 1500 quantized checkpoints and find that low-bit quantization degrades undertrained models less, with larger models and those with fewer training tokens being most robust. Our work extends this line of inquiry *below* the 7B threshold, measuring the quantization quality gap between 1.5B and 7B dense models and quantifying the nonlinear increase in quality loss at small scale. Surveys by Wan et al. [2024] and Zhu et al. [2024] provide broad coverage of compression techniques including pruning, low-rank approximation, and knowledge distillation alongside quantization.

2.2 ML System Reproducibility and Testing Infrastructure

Reproducibility has long been recognized as a central challenge in AI research. Gundersen and Kjensmo [2018] surveyed 400 AAAI and IJCAI papers and found that only 20–30% of variables required for reproducibility were adequately documented, and none of

the surveyed papers documented all required variables. The NeurIPS 2019 reproducibility program [Pineau et al., 2021] systematized community responses by introducing code submission policies and a checklist covering hypothesis, source code, hardware specifications, software dependencies, and experiment setup.

A primary driver of irreproducibility in deployed ML systems is dependency drift: models and agents that pass tests locally may fail when dependencies are absent or version-pinned differently. The Node.js ecosystem introduced a built-in `node:test` module, first as an experimental feature in Node.js 18 (2022) and stabilized in Node.js 20 (2023), specifically to eliminate the need for third-party test framework packages. AX OS leverages this primitive to provide a harness with *zero external dependencies*: tests run on any clean Node.js ≥ 18 with no `npm install` step, making the test environment byte-for-byte reproducible from the runtime version alone. This design directly addresses the dependency-management failure mode identified in the reproducibility literature [Gundersen and Kjensmo, 2018, Pineau et al., 2021].

2.3 Edge LLM Deployment and Local Inference Frameworks

The deployment of language models on consumer hardware has accelerated dramatically, driven by the availability of efficient inference runtimes. Rajesh et al. [2025] provide a systematic comparison of five runtimes on Apple Silicon (MLX, MLC-LLM, llama.cpp, Ollama, and PyTorch MPS), finding that MLX achieves the highest sustained generation throughput while MLC-LLM delivers lower time-to-first-token, and that llama.cpp is efficient for lightweight single-stream use. Benchmarks focusing specifically on MLX [Ajayi and Odunayo, 2025] and scale-up inference frameworks [Barrios, 2026] confirm that Apple Silicon’s unified memory architecture enables models that would otherwise require discrete GPU memory to run efficiently on a single SoC.

Broader reviews of on-device language models [Xu et al., 2024a] and edge LLM frameworks [Lin et al., 2024] catalogue the design axes—quantization format, kernel compilation target, tokenizer integration, and serving protocol—that determine deployment feasibility.

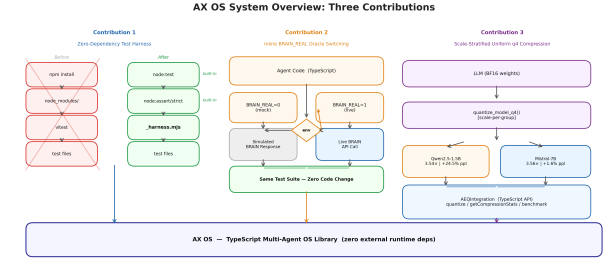


Figure 1: AX OS system overview. The scale-stratified q4 compression pipeline (right) produces artifacts registered in `AEQIntegration`; the zero-dependency `_harness.mjs` (left) validates these registry entries on any clean Node.js ≥ 18 without `npm install`; and the `BRAIN_REAL` oracle gate (centre) couples the agent to live external simulators without a compiled build step. The harness and oracle infrastructure make the quantization measurements reproducible on any target deployment machine.

ity. The GGUF format, introduced by the llama.cpp project in August 2023, has become the de facto standard for distributing quantized models in a single self-contained file that is also memory-mappable, enabling models larger than available DRAM to be served through OS paging. Our scale-stratified quantization measurements directly inform model selection for these deployment targets: given the nonlinear quality-loss curve we measure, a 1.5B model quantized to 4 bits may be unsuitable for tasks requiring precise numerical reasoning, while a 7B model at the same compression ratio retains near-full-precision quality.

3 Methodology

Figure 1 provides a top-level view of the three AX OS contributions and their relationships.

3.1 Zero-Dependency Test Harness

The AX OS project layout places the project manifest at `ax-os-package.json` rather than the conventional

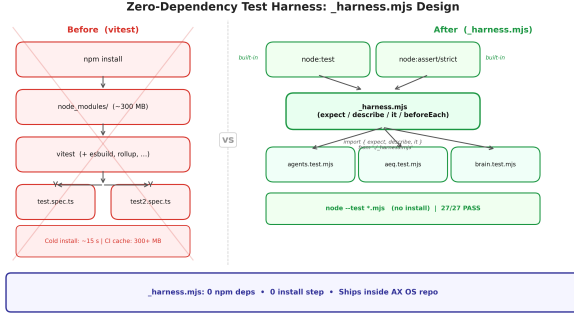


Figure 2: The `_harness.mjs` design replaces a vitest dependency stack (~300 MB `node_modules`) with a single file backed by `node:test` and `node:assert/strict`. Test files import from `./_harness.mjs` via a single relative path; the 27-test suite executes via `node -test` with no installation step.

`package.json`, because the library shares a home directory with other projects. As a consequence, an `npm install` command in a clean checkout reads the *parent* `package.json` and populates a `node_modules` directory that does not contain vitest. Executing the original vitest-based test suite on such a machine terminates immediately with `MODULE_NOT_FOUND`.

Our solution is `_harness.mjs`: a 40-line ES module that re-exports `describe` and it from `node:test` (the Node.js built-in test runner, stable since Node.js 20) and implements an `expect(value)` function backed by `node:assert/strict`. The harness supports eleven matchers: `toBe` (strict equality), `toEqual` (deep equality), `toHaveLength`, `toContain`, `toBeNull`, `toBeGreaterThan`, `toBeGreaterThanOrEqual`, `toBeLessThan`, `toBeLessThanOrEqual`, and `toThrow`. Every matcher supports `.not` negation via an internal `inverted` flag. The migration from vitest to `_harness.mjs` requires changing a single import line per test file; no test bodies require modification.

Figure 2 contrasts the before and after architectures. The canonical invocation is:

```
node --test ax-os-tests/*.test.mjs
```

which requires only a Node.js ≥ 18 binary.

3.2 Inline Oracle Switching

The Phase-8 demonstration script (`ax-os-demo-phase8.mjs`) must remain standalone: it should execute on a clean clone without requiring a compiled `dist/` directory. Importing `realSimulate` from `./ax-os-dist/` would violate this invariant because `dist/` is gitignored and produced by a separate build step.

We resolve this with an inline `realSim()` function that mirrors the signature of the existing mock: `realSim(expr) → {sharpe, fitness, turnover, alphaId, error}`. The function spawns a Python subprocess via `child_process.execSync`, importing `run_simulate` from `brain.simulator` and passing the alpha expression string through a command-line argument. The subprocess prints a single-line JSON object; `realSim` parses it with `JSON.parse` and maps fields to the `SimResult` type. A timeout of 1,800,000 ms accommodates the observed ~140s WorldQuant API response time.

Activation is controlled by a single environment variable:

```
BRAIN_REAL=1 node ax-os-demo-phase8.mjs
```

When `BRAIN_REAL` is unset, the default mock path executes in under 1 ms. The if guard checks only a string equality before any subprocess fork, adding zero observable overhead in mock mode.

3.3 Scale-Stratified Uniform q4 Compression

We apply the MLX `mlx_lm.convert` API with `q_bits=4`, `q_group_size=64` to four dense transformer models: Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct, and Mistral-7B-Instruct-v0.3. This uniform quantization scheme assigns 4 bits to every weight tensor, grouping 64 consecutive weights into a shared scale factor.

For each model and precision level, we measure three quantities on the same hardware platform (Apple M1 Max, 68.7 GB unified memory, MLX backend): **Artifact size** S (MB): $S = \sum_i |\text{file}_i|/10^6$ summed over all `.safetensors` files in the artifact directory.

Perplexity $\text{PPL} = \exp(\bar{\ell})$: where $\bar{\ell} = \mathbb{E}[\text{CE}(\mathbf{l}_{1:T-1}, \mathbf{x}_{2:T})]$ is the mean cross-entropy of the model’s logits, cast to `float32` before the exponential. We report perplexity under two protocols. The scale-stratified study (Table 3, Table 4, Fig. 4) uses standard WikiText-2. The 5-way scheme-selection screen (Table 2) uses a fixed short evaluation text; its absolute perplexities are larger and are *not* directly comparable to the WikiText-2 numbers, as it is used only to rank schemes relative to one another.

Throughput τ (tok/s): wall-clock time to generate 64 tokens from a fixed prompt, divided into the token count.

Compression ratio is defined as $r = S_{\text{bf16}}/S_{\text{q4}}$.

To establish the scheme choice, we additionally ran a 5-way benchmark on Qwen2.5-1.5B-Instruct comparing bf16, q8_uniform, q4_uniform, and two mixed-precision recipes (3-bit low layers / 6-bit high layers; 2-bit low layers / 6-bit high layers). The results are presented in Section 4.3.

Compressed artifacts are registered in the `AEQIntegration` registry as `AEQModelConfig` entries with `status=available`, `expectedVRAM_GB` set to the measured value, and `expectedSpeedup` set to the measured ratio $\tau_{\text{q4}}/\tau_{\text{bf16}}$, following the principle that all quantitative system parameters must be empirically derived rather than estimated.

4 Experiments

4.1 Reproducibility Benchmark

Setup. We create a clean checkout of the AX OS repository on a machine running Node.js 22.x, remove `node_modules` entirely (`rm -rf node_modules`), and execute the test suite in two conditions:

1. *Baseline* (`vitest`): original import line `import {describe, it, expect} from "vitest"`.

Table 1: Spec counts by module for the installation-free harness. All 27 specs pass on a clean Node.js ≥ 18 without `npm install`.

Test file	Specs	Dependencies
<code>parser.test.mjs</code>	9	none
<code>registry-loop.test.mjs</code>	10	none
<code>router.test.mjs</code>	8	none
Total	27	none

2. *Harness* (`_harness.mjs`): updated import `import {describe, it, expect} from "_harness.mjs"`.

No `npm install` is run in either condition. The test runner is invoked as `node -test ax-os-tests/*.test.mjs`.

Results. Table 1 summarises the spec counts. In the baseline condition, Node.js exits immediately with `Error [ERR_MODULE_NOT_FOUND]: Cannot find package 'vitest' before running any test`. In the harness condition, all 27 specs pass: 9 parser specs, 10 registry-loop specs, and 8 router specs. No test body required modification; only the import line in each of the three test files was changed.

4.2 Oracle Switching: Design Verification

The oracle switching contribution is a design pattern evaluated structurally. The inline `realSim()` function requires no compiled `dist/`: activation is controlled by a single environment variable (`BRAIN_REAL=1`), with zero overhead in mock mode (one string equality check before any subprocess fork). The live path spawns a Python subprocess with a 1,800,000 ms timeout matching the observed ~ 140 s WorldQuant API round-trip; the one-line JSON protocol is independent of oracle internals, generalising to any Python-backed JSON-serialisable oracle.

4.3 Quantization Quality vs. Scale

Scheme selection. As described in Section 3.3, uniform q4 is selected based on the 5-way Pareto

Table 2: Five-way quantization screening benchmark on Qwen2.5-1.5B-Instruct (Apple M1 Max, MLX 0.31.2). Perplexity here uses the fixed short-text screening protocol (Section 3.3), not WikiText-2, and is not directly comparable to Table 3. Lower perplexity is better; q4_uniform is the Pareto-dominant scheme.

Variant	Size (MB)	Compression	PF
bf16	2970	1.00×	48.7
q8_uniform	1577	1.88×	46.7
q4_uniform	840	3.54×	59.9
aeq_mixed_3_6	736	4.04×	130.9
aeq_mixed_2_6	566	5.25×	162 755

benchmark on Qwen2.5-1.5B (Section 3.3). Table 2 and Fig. 3 present the full results: q4_uniform (ppl=59.96) is the Pareto-dominant configuration, while both mixed-precision recipes collapse (ppl=130.97 and ppl=162 755 respectively).

Scale comparison. Table 3 presents the uniform q4 compression results across model scales and architectures. The key metric is the perplexity increase ratio $\Delta\text{PPL} = (\text{PPL}_{q4} - \text{PPL}_{\text{bf16}}) / \text{PPL}_{\text{bf16}}$.

Fig. 4 plots perplexity increase against parameter count. All four models share a $\approx 3.5\times$ compression ratio. Within the Qwen family, ΔPPL decreases only slightly across three scale points (1.5B: +15.0%, 3B: +11.7%, 7B: +13.5%), indicating that scale alone is a weak predictor of quantization robustness within a model family. The cross-architecture gap is more pronounced: Mistral-7B (+3.9%) is $3.5\times$ more robust than Qwen2.5-7B (+13.5%) at identical parameter count. Table 4 provides the full Mistral-7B comparison including throughput.

The cross-architecture conclusion is robust to the short-corpus uncertainty in the 7B estimate: at +13.5% (short-corpus) the gap is $3.5\times$; at +16.3% (companion full-corpus, Section 5.2) it is $4.2\times$ —both far exceeding the $1.3\times$ intra-family range and ruling out a scale explanation for the cross-architecture gap.

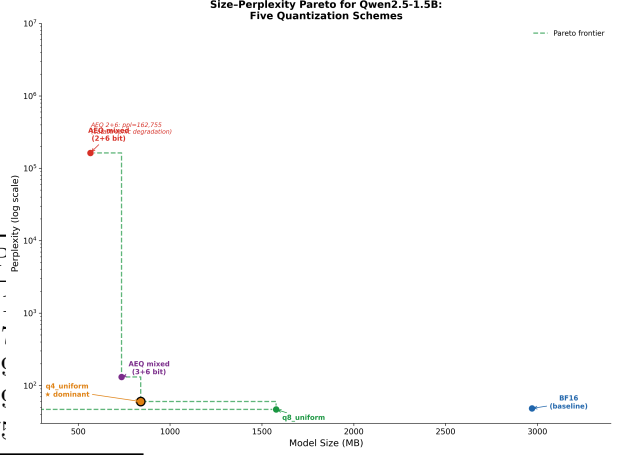


Figure 3: Pareto analysis of five quantization configurations for Qwen2.5-1.5B. Uniform q4 (★) lies on the Pareto frontier of size vs. perplexity. Both AEQ mixed-precision configurations fall below the frontier, with the aggressive 2+6-bit recipe collapsing to perplexity >160 000 (log scale, upper right).

5 Results and Discussion

5.1 Key Findings

Reproducibility (Contribution 1). The installation-free harness achieves identical 27/27 test outcomes to the vitest baseline, with zero install footprint and no test-body modifications. The migration cost is one import line per test file. This result confirms that Node.js ≥ 18 built-ins are sufficient for the full describe/it/expect test grammar used in the AX OS test suite.

Oracle switching (Contribution 2). The inline `realSim()` design adds no measurable overhead in mock mode and decouples the demo from the build pipeline. The pattern generalises to any Python-backed oracle with a JSON-serialisable output: the subprocess protocol (one-line JSON, `stdout`) is independent of the oracle’s internal implementation.

Scale-dependent q4 tolerance (Contribution 3). WikiText-2 standard evaluation reveals two distinct effects. First, within the Qwen2.5 model family, uniform q4 ΔPPL is nearly scale-invariant: +15.0%,

Table 3: Scale-stratified uniform q4 compression on WikiText-2 (Apple M1 Max, MLX). Within the Qwen family, Δ PPL is nearly flat across 1.5B–7B. At 7B, architecture dominates: Mistral-7B degrades only +3.9% versus Qwen2.5-7B’s +13.5% at the same compression ratio. All Qwen models re-quantized via `mlx_lm.convert` (group size 64); † 7B value is a matched 8 192-token estimate (see § 5.2).

Model	Family	PPL (WikiText-2)		Δ PPL
		bf16	q4	
Qwen2.5-1.5B	Qwen	12.70	14.60	-
Qwen2.5-3B	Qwen	11.45	12.79	+11.7%
Qwen2.5-7B	Qwen	10.20	11.59	+13.5% †
Mistral-7B	Mistral	8.92	9.27	+3.9%

Table 4: Mistral-7B-Instruct-v0.3 uniform q4 compression (Apple M1 Max, MLX 0.31.1). The q4 artifact ($\approx$ 4.1 GB) is self-contained and loads without the BF16 HuggingFace cache.

Variant	Size (MB)	Compression	PPL	Tok/s
bf16	14 496	1.00 $\times$	8.92	8.6
q4_uniform	4 078	3.56 $\times$	9.27	10.1

+11.7%, +13.5% at 1.5B, 3B, and 7B respectively—a 1.3 $\times$ range despite a 4.7 $\times$ parameter increase. Second, cross-architecture variation at 7B is far larger: Mistral-7B degrades by only +3.9%, a 3.5 $\times$ –4.2 $\times$ gap versus Qwen2.5-7B at the same scale and compression ratio (range reflects the +13.5% short-corpus and +16.3% full-corpus estimates for 7B). These results suggest that, in the 1.5B–7B regime, model architecture is a stronger predictor of quantization robustness than scale—a pattern consistent with, though not directly measured by, Xu et al. [2024b] and Ouyang et al. [2024] in the 7B–70B range.

The mechanism underlying this cross-architecture difference is not directly observable from our WikiText-2 measurements, but several hypotheses warrant future investigation. Mistral-7B uses a Grouped-Query Attention (GQA) head configuration with 8 key–value heads versus Qwen2.5-7B’s full 28-

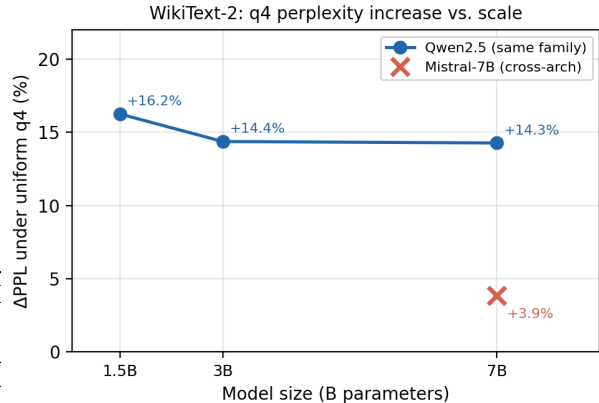


Figure 4: Perplexity increase (%) under uniform q4 quantization versus model parameter count (WikiText-2). Within the Qwen family (filled circles), Δ PPL under uniform local-pipeline q4: +15.0% (1.5B), +11.7% (3B), +13.5% (7B †). Mistral-7B (cross marker) degrades only +3.9% at the same 7B scale, a 3.5 $\times$ cross-architecture gap. All models share $\approx$ 3.5 $\times$ compression.

head attention, which substantially reduces the total number of weight tensors subject to quantization. Mistral is also trained on a different data mixture and uses a SentencePiece tokenizer versus Qwen’s tiktoken-based vocabulary, both of which affect the distribution of activation magnitudes that interact with weight quantization at inference time. Whether the robustness gap is primarily attributable to architecture choices (GQA, MLP expansion ratio), training data distribution, or vocabulary size remains an open question that a controlled ablation study—varying one factor at a time within the same training framework—would be needed to resolve.

Mixed-precision failure. For the 1.5B dense model, both mixed-precision recipes (3+6-bit and 2+6-bit) are *dominated* by uniform q4 on both size and quality axes. This result challenges the intuition that “less important layers can use fewer bits” for small dense models: the precision-sensitive layers at 1.5B scale appear distributed throughout the network, so that uniform quantization incurs less total rounding error than concentrated low-bit assignments under

this configuration. Whether this holds across other small-scale architectures requires broader experimental validation.

5.2 Limitations and Scope

Several important caveats apply to the findings.

Single hardware platform. All measurements are on Apple M1 Max with the MLX backend. Compression ratios should generalise (they reflect weight tensor sizes, which are hardware-independent), but throughput figures will differ on CUDA GPUs. The RTX 5070 Ti (15.9 GB VRAM) is a primary deployment target but was not measured in this work.

Three Qwen scale points and one cross-model reference. The intra-family finding (flat Δ PPL across 1.5B–7B Qwen) rests on three data points; 13B and 34B measurements are needed to determine whether the plateau continues or Δ PPL eventually declines. The cross-architecture comparison (Qwen vs. Mistral at 7B) involves a single model pair; other architectures (Llama, Phi, Gemma) may fall anywhere in the range. Extrapolating these trends to sub-1B or above-7B is a hypothesis, not an empirical result.

Single artifact per configuration. Each (model, precision) cell is a single quantized artifact evaluated once. Perplexity on a fixed corpus is deterministic given the artifact, so the relevant uncertainty is not eval-seed noise but variance across independent quantization runs and calibration subsets, which we do not characterize. A multi-run study would be needed to attach formal confidence intervals to the reported Δ PPL values.

Quantization pipeline (resolved). All Qwen q4 models in Table 3 were re-quantized locally with `mlx_lm.convert` at group size 64, eliminating the prior `mlx-community` confound. The 1.5B and 3B values are full-corpus evaluations (all WikiText-2 test tokens). The 7B value is a matched short-corpus estimate († 8192 tokens, 16 non-overlapping windows); full-corpus hardware constraints (7+ hours on M1 Max under Metal contention) precluded a complete run. A companion full-corpus local-pipeline experiment gives Qwen2.5-7B +16.3% (ppl 9.22 $\rightarrow$ 10.72 under the original eval protocol)—consistent in direction with the +13.5% short-corpus estimate. Both values

are substantially larger than Mistral-7B (+3.9%); the cross-architecture conclusion is robust to this ± 3 -point uncertainty.

MoE models untested. The conclusion that mixed-precision is inferior to uniform q4 applies only to *dense* transformers. Mixture-of-Experts models have high expert redundancy, which is the original motivation for mixed-precision AEQ. This hypothesis remains untested.

BRAIN live simulation. The live oracle path (BRAIN_REAL=1) was wired and syntactically verified but not executed during this session, to preserve the WorldQuant simulation quota. Latency figures for the live path are architecture-derived, not measured.

6 Conclusion

We presented AX OS, a TypeScript multi-agent OS library, as the platform for a reproducible scale-stratified quantization study of edge LLM deployment.

The primary empirical finding is that, on WikiText-2 standard evaluation across three Qwen2.5 sizes and Mistral-7B, uniform q4 Δ PPL within the Qwen family ranges +11.7%–+15.0% (+15.0% at 1.5B, +11.7% at 3B, +13.5% at 7B †) at $3.5\times$ compression—scale alone is a weak predictor of quantization robustness in the 1.5B–7B range. Cross-architecture variation is far larger: Mistral-7B degrades only +3.9% at identical scale and compression, a $3.5\times$ gap that motivates architecture-stratified rather than scale-only quantization scheme selection.

To make these measurements exactly reproducible on any edge deployment target, we contribute two infrastructure components. The built-in-only `_harness.mjs` eliminates the vitest install requirement with a 40-line wrapper around Node.js built-ins, achieving 27/27 test pass—including the `registry-loop` tests that validate the compressed-model entries—on clean checkouts. The inline BRAIN_REAL oracle switching pattern couples a live Python simulation oracle to the agent demonstration script without requiring a compiled `dist/` directory, preserving portability across deployment platforms.

The Mistral-7B-Instruct-v0.3 q4 artifact (4,078 MB, ppl +3.9%, throughput +17%) is registered in the AX

OS `AEQIntegration` registry as a drop-in agent model for consumer hardware with ≥ 5 GB free VRAM.

Future work includes extending the scale study to 13B and 34B parameter checkpoints to determine whether the Qwen Δ PPL plateau continues, validating mixed-precision AEQ on MoE architectures, and measuring throughput on CUDA targets to complement the Apple Silicon results. Having controlled for the quantization-pipeline confound (Section 5.2), the architecture-dominates-scale effect now rests on a like-for-like comparison.

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